# Incorporating the Luce-Krantz threshold model into the cultural consensus theory for ordinal categorical data: A simulation study

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# Outline

Introduction

General Condorcet-Luce-Krantz Model

Simulation Results

General Discussion

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# Traditional Test Theory




Test score

Competency

But,
if we do not have the answer key in prior?

# Cultural Consensus Theory

aka a "Test Theory Without an Answer Key"

- A cognitive-based methodology developed by Batchelder and colleagues in the mid-1980s (Batchelder & Romney, 1986; Batchelder & Romney, 1988; Romney et al., 1986, 1987)
- Researchers design a set of relevant items (or questions) for informants who have different knowledge about their common or shared culture.
- However, the researchers do not know the answers to items a priori.

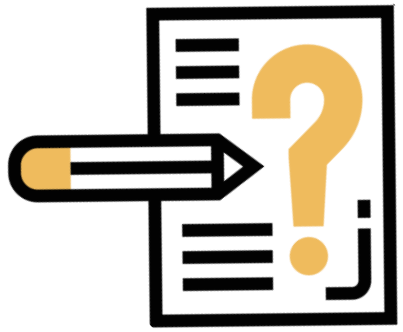
## Goals

- To estimate the consensus **answers** to the items shared by the group
- Response characteristics: informant's competency and bias, item's difficulty

# $N$ informants respond to $M$ items

Consensus truth: $Z_k \in \{0,1\}$; Response format: $X_{ik} \in \{1, \ldots, J\}$

E.g., 5 points Likert scale



Disagree Agree

1 2 3 4 5

| | $Item_1$ | $Item_2$ | $Item_3$ | ... | $Item_k$ | ... | $Item_M$ |
|---|---|---|---|---|---|---|---|
| $Info_1$ | 2 | 5 | 4 | ... | ... | ... | 4 |
| $Info_2$ | 3 | 1 | 1 | | | | 5 |
| $Info_3$ | 2 | 4 | 2 | | | | 4 |
| ⋮ | ⋮ | | | ⋱ | | | ⋮ |
| $Info_i$ | ⋮ | | | | $x_{ik}$ | | ⋮ |
| ⋮ | ⋮ | | | | | ⋱ | ⋮ |
| $Info_N$ | 2 | 4 | 3 | ... | ... | ... | 5 |

- Item's consensus truth/answer & difficulty
- Informant's competency & response bias
- Clustering of the informants

# Outline

Introduction

**General Condorcet-Luce-Krantz Model**

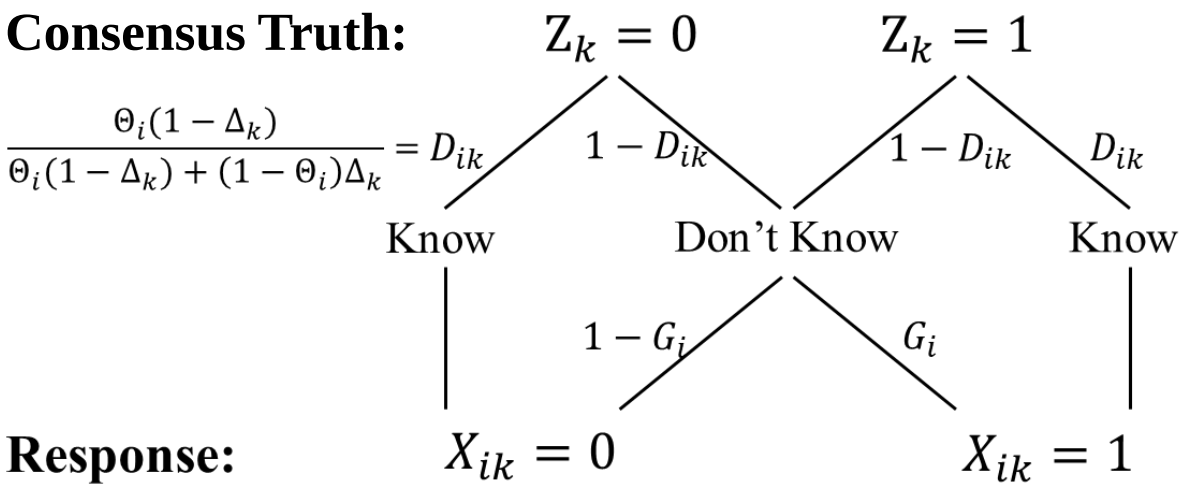
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# General Condorcet-Luce-Krantz Model (GCLK)

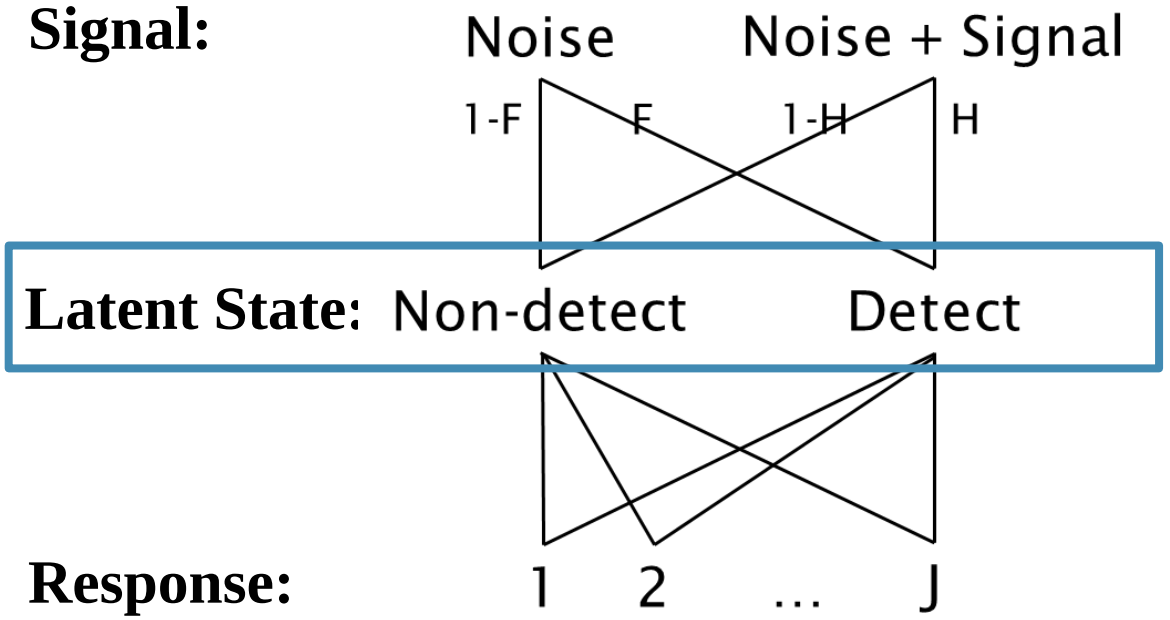
## General Condorcet Model

(Batchelder & Anders, 2012)

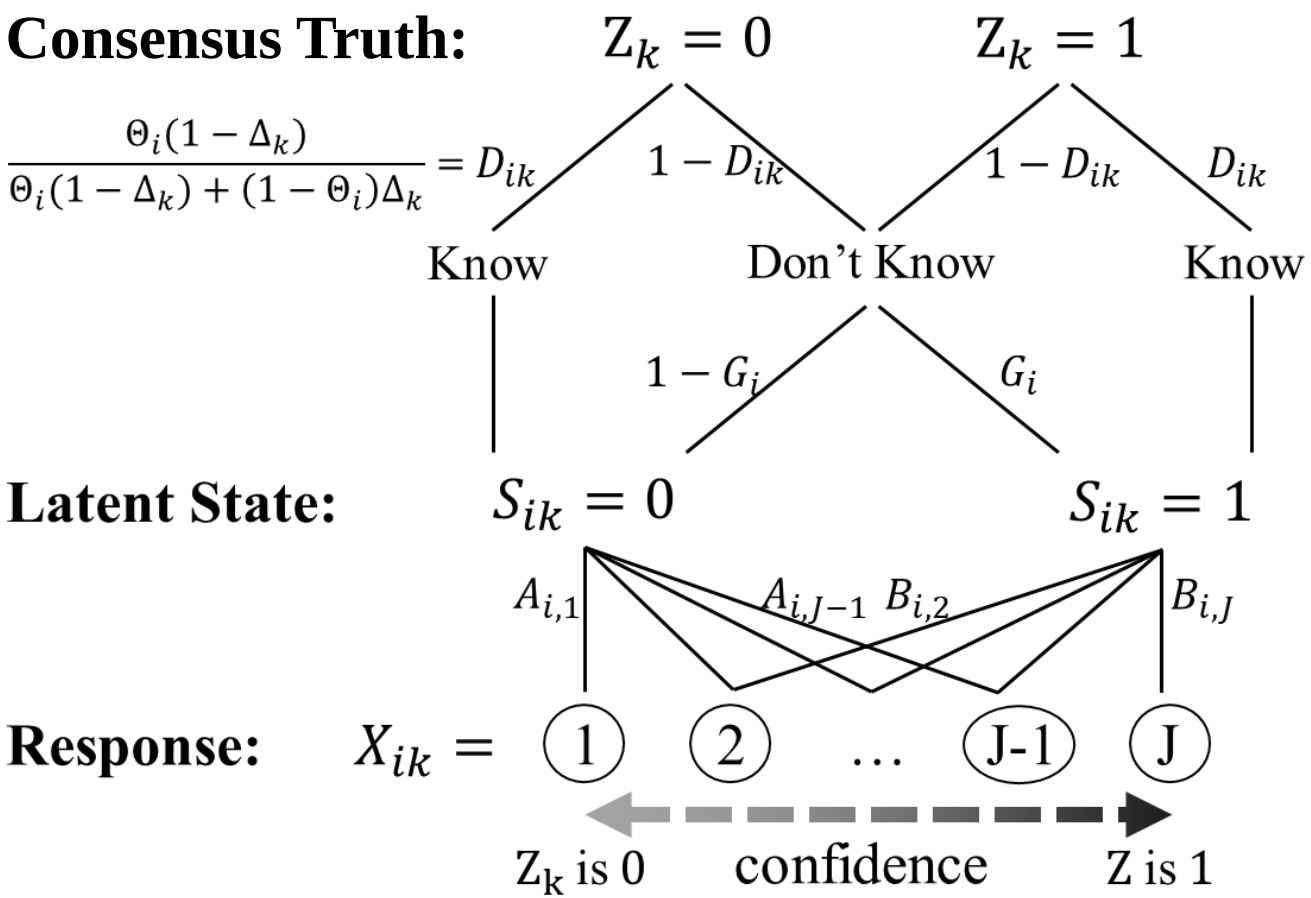



## The two-state low-threshold model with confidence ratings

(Luce, 1963; Krantz, 1969)



# General Condorcet-Luce-Krantz Model (GCLK)




- Informant: $i = 1, \ldots, N$;
- Item: $k = 1, \ldots, M$
- Consensus truth: $Z_{ik} = \{0, 1\}$
- Latent state: $S_{ik} = \{0, 1\}$
- Response: $X_{ik} \in \{1, \ldots, J\}$
- Item difficulty: $\Delta_k$
- Informant's competency: $\theta_i$
- Informant's guessing bias: $G_i$
- Informant's response criteria (Decision biases):`$A_i$, $B_i$

# Hierarchical Bayesian Model of the GCLK

$$Z_{\nu k} \sim \text{Bernoulli}(p_{Z_\nu})$$

$$\Theta_i \sim \text{Beta}\left(\mu_{\Theta_{\Omega_i}}\tau_{\Theta_{\Omega_i}}, \left(1-\mu_{\Theta_{\Omega_i}}\right)\tau_{\Theta_{\Omega_i}}\right)$$

$$\Delta_{\nu k} \sim \text{Beta}(\mu_{\Delta_\nu}\tau_{\Delta_\nu}, (1-\mu_{\Delta_\nu})\tau_{\Delta_\nu})$$

$$G_i \sim \text{Beta}\left(\mu_{G_{\Omega_i}}\tau_{G_{\Omega_i}}, \left(1-\mu_{G_{\Omega_i}}\right)\tau_{G_{\Omega_i}}\right)$$

$$\boldsymbol{A}_i \sim \text{Dirichlet}\left(\boldsymbol{\alpha}_{A_{\Omega_i}}\right)$$

$$\boldsymbol{B}_i \sim \text{Dirichlet}\left(\boldsymbol{\alpha}_{B_{\Omega_i}}\right)$$

$$\Omega_i \sim \text{Categorical}(\boldsymbol{p}_\Omega)$$

$$p_{Z_\nu} \sim \text{Uniform}(0,1)$$

$$\mu_{\theta_{\Omega_i}} \sim \text{Beta}(\alpha_{\mu_\theta}, \alpha_{\mu_\theta}), \quad \alpha_{\mu_\theta} = 2$$

$$\tau_{\theta_{\Omega_i}} \sim \text{Gamma}(\mu_{\tau_\theta}^2/\sigma_\theta^2, \mu_{\tau_\theta}/\sigma_{\tau_\theta}^2), \quad \mu_{\tau_\theta} = 10, \quad \sigma_{\tau_\theta} = 10$$

$$\mu_{\Delta_\nu} = 0.5$$

$$\tau_{\Delta_\nu} \sim \text{Gamma}(\mu_{\tau_\Delta}^2/\sigma_{\tau_\Delta}^2, \mu_{\tau_\Delta}/\sigma_{\tau_\Delta}^2), \quad \mu_{\tau_\Delta} = 10, \quad \sigma_{\tau_\Delta} = 10$$

$$\mu_{G_{\Omega_i}} = 0.5$$

$$\tau_{G_{\Omega_i}} \sim \text{Gamma}(\mu_{\tau_G}^2/\sigma_{\tau_G}^2, \mu_{\tau_G}/\sigma_{\tau_G}^2), \quad \mu_{\tau_G} = 10, \quad \sigma_{\tau_G} = 10$$

$$\boldsymbol{\alpha}_{A_{\Omega_i}} = (1, \ldots, 1, 0)^\top$$

$$\boldsymbol{\alpha}_{B_{\Omega_i}} = (0, 1, \ldots, 1)^\top$$

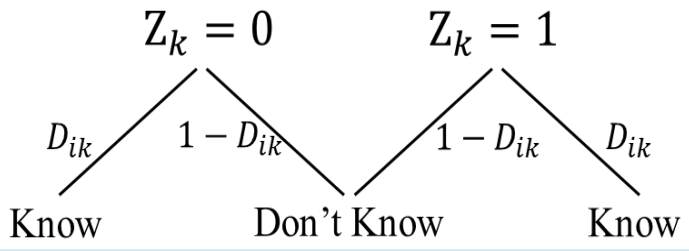
$$\boldsymbol{p}_\Omega \sim \textit{Dirichlet}(\mathbf{L}), \quad \mathbf{L} = (1, \ldots, 1)^\top$$

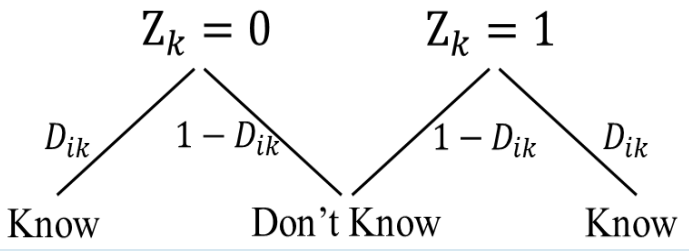
# Multicultural GCLK

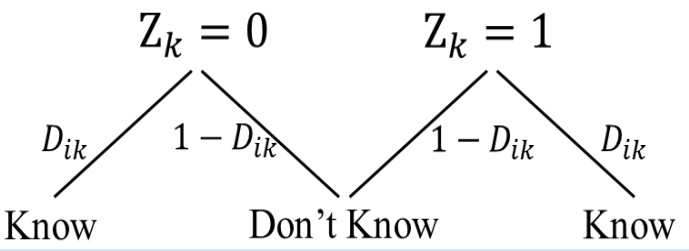


$$\boldsymbol{\Omega} = (\boldsymbol{\Omega}_1, \ldots, \boldsymbol{\Omega}_M)^\top \in \{1, \ldots, V\}^M$$

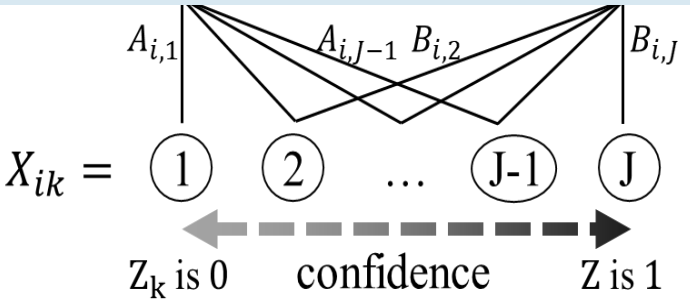
$$\boldsymbol{Z_{v=1}} = (Z_{11}, \ldots, Z_{1M})^\top \qquad \boldsymbol{Z_{v=2}} = (Z_{21}, \ldots, Z_{2M})^\top \qquad \boldsymbol{Z_{v=V}} = (Z_{V1}, \ldots, Z_{VM})^\top$$

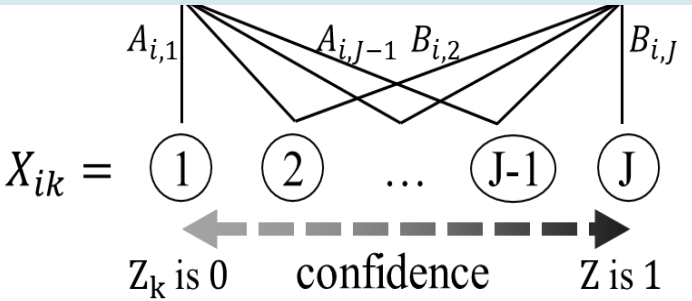


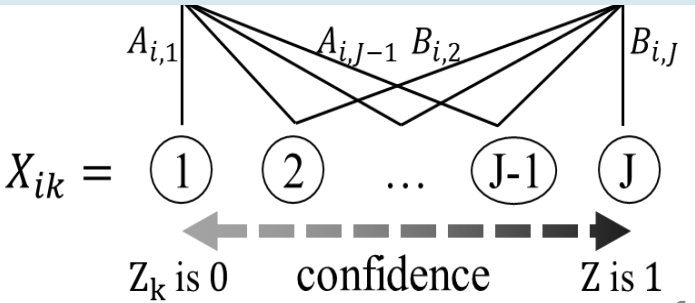




**How many cultures $V$ should we choose for the GCLK when applying it to observed data?**



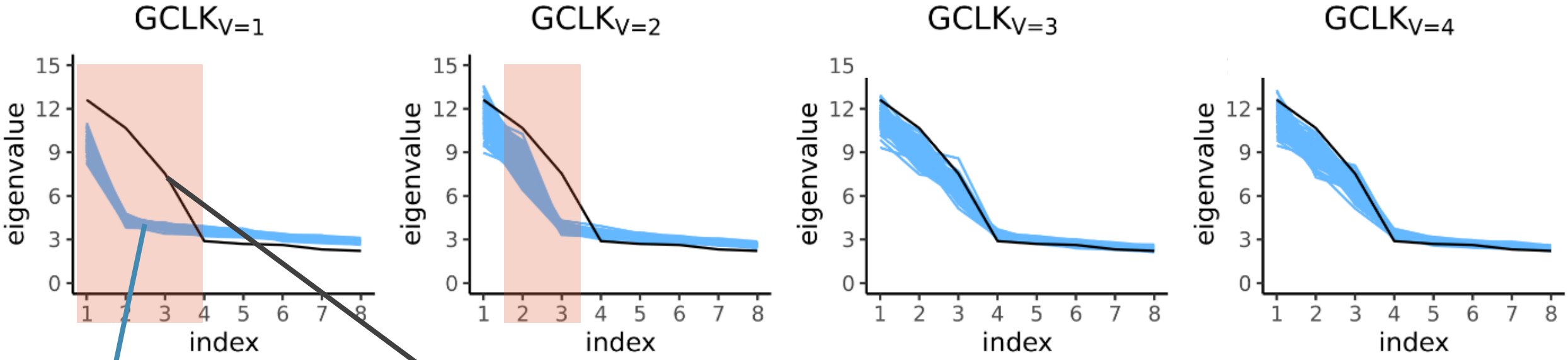



# Posterior Predictive Check & DIC

Four GCLK with $V = 1, \ldots, 4$ fitted to the three-culture data ($SIM_{V=3}$)

1. posterior predictive check
2. favor a model with the smallest deviance information criterion (DIC) value



# Outline

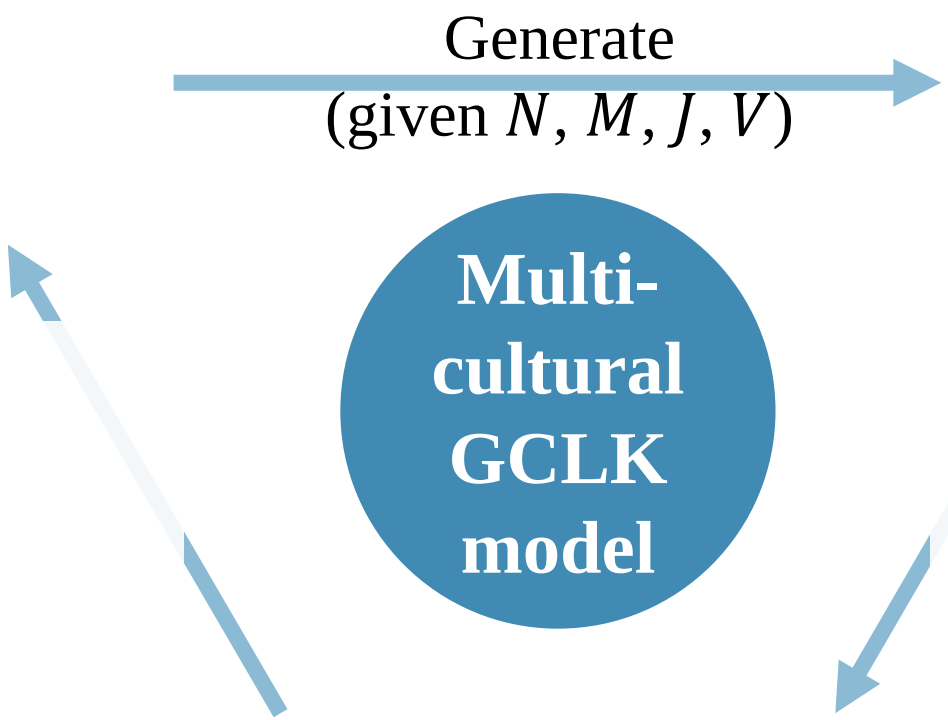
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**Simulation Results**

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# Procedure of the Simulation Study



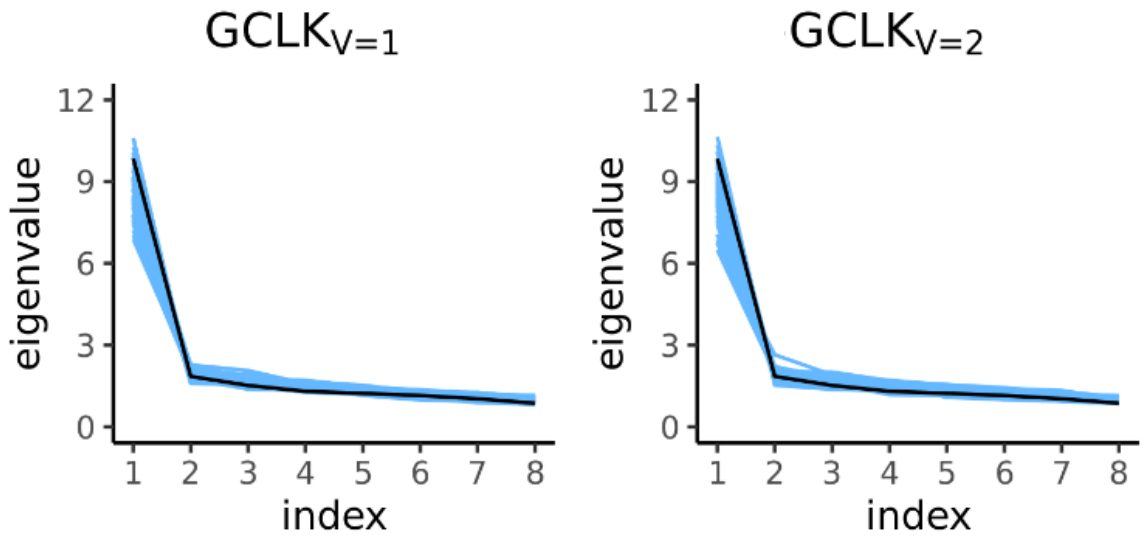

Parameter recovery
- model selection
- posterior mean
- 95% HDI

Markov Chain Monte-Carlo
- 6 chains
- 6000 iterations
- 1000 iter. as burn-in
- scrutinize label-switching and mixed-number phenomena

# Simulation results ($N = 25, M = 50, J = 5, V = 1$)

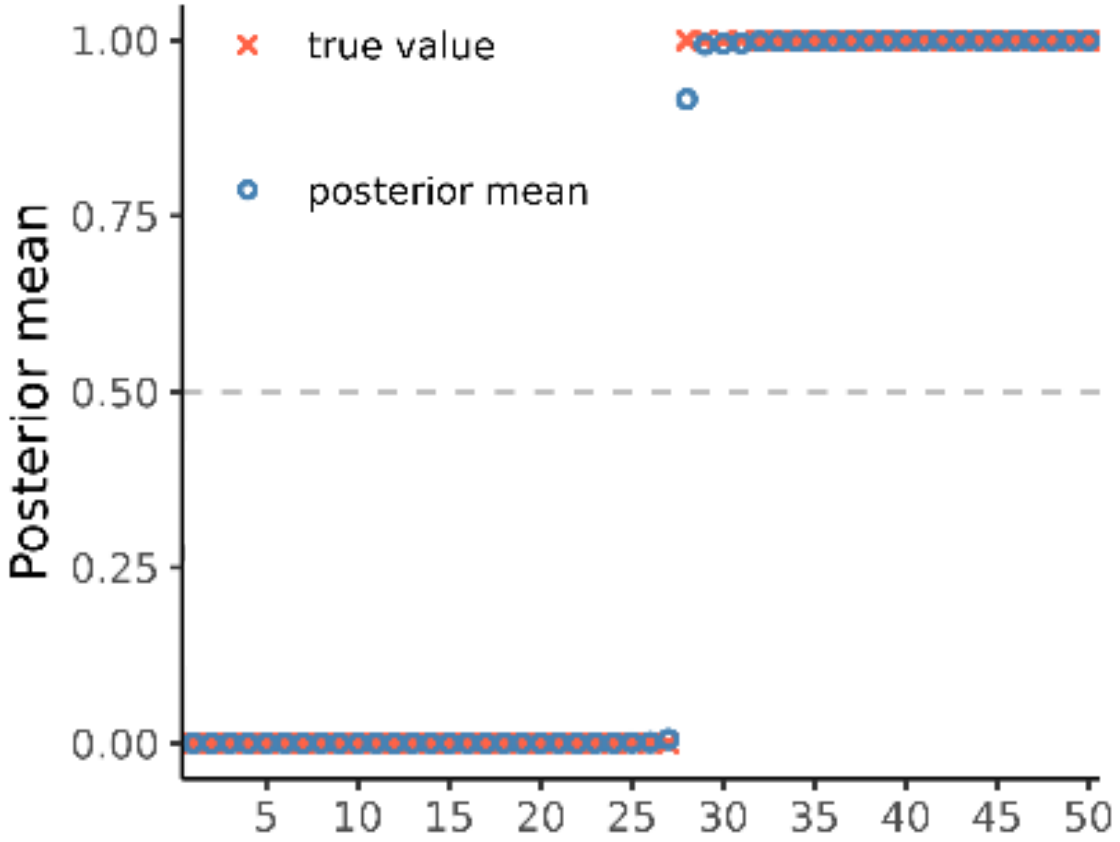
The estimate cultural number and the DIC

Consensus truth: $Z$

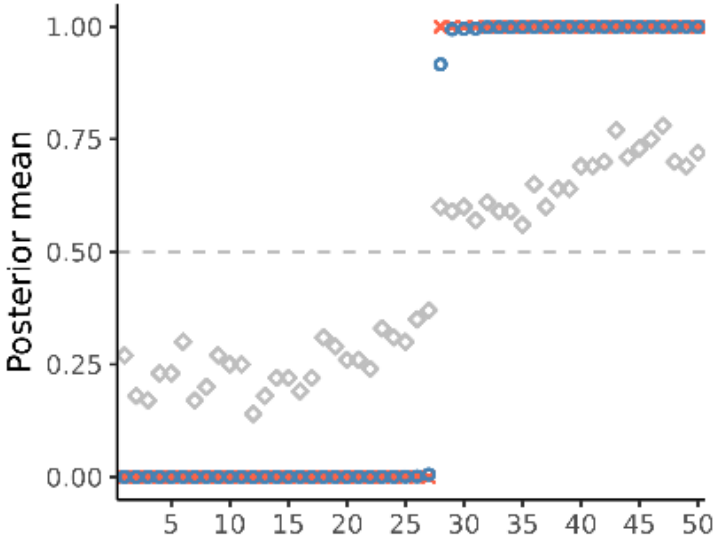



DIC = 3552.31

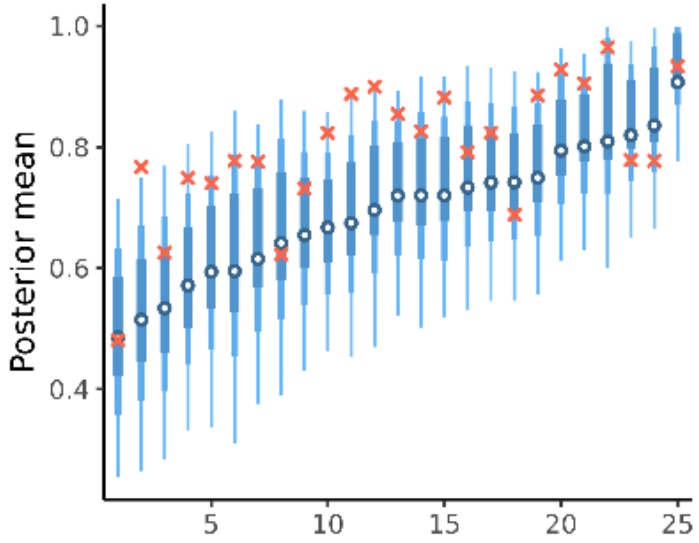
DIC = 3558.89



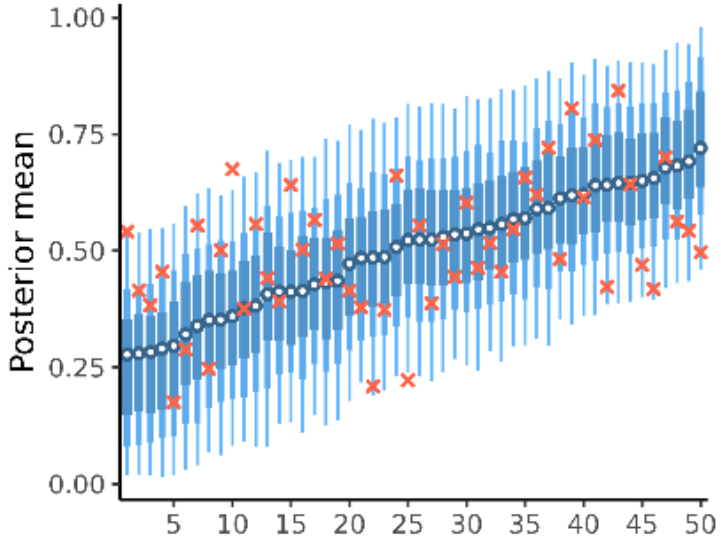
# Simulation results ($N = 25, M = 50, J = 5, V = 1$)



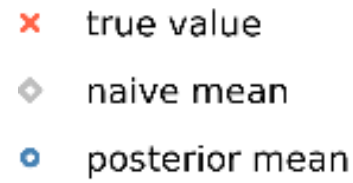

Consensus truth: $Z$

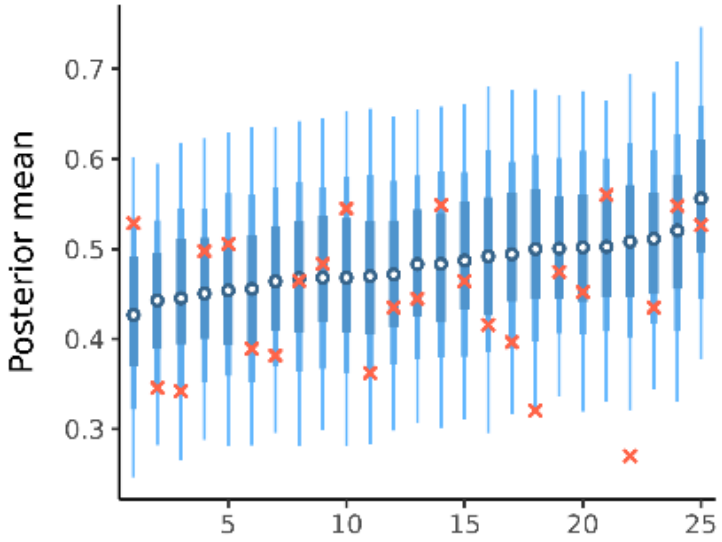



Competency: $\Theta$

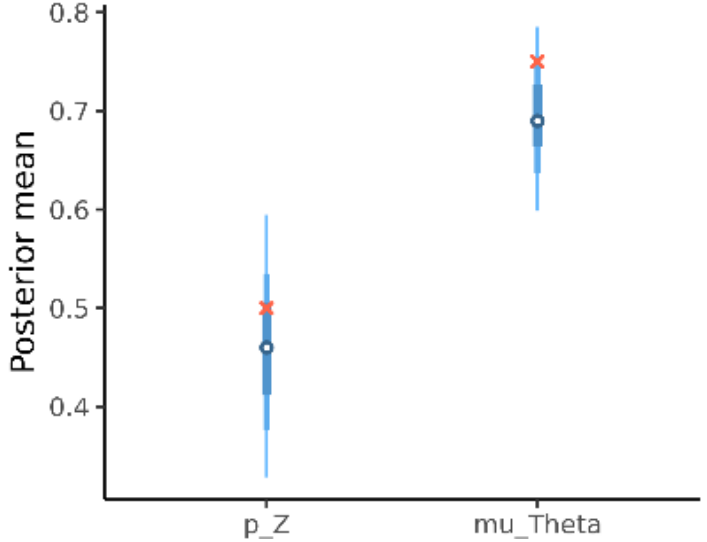



Difficulty: $\Delta$

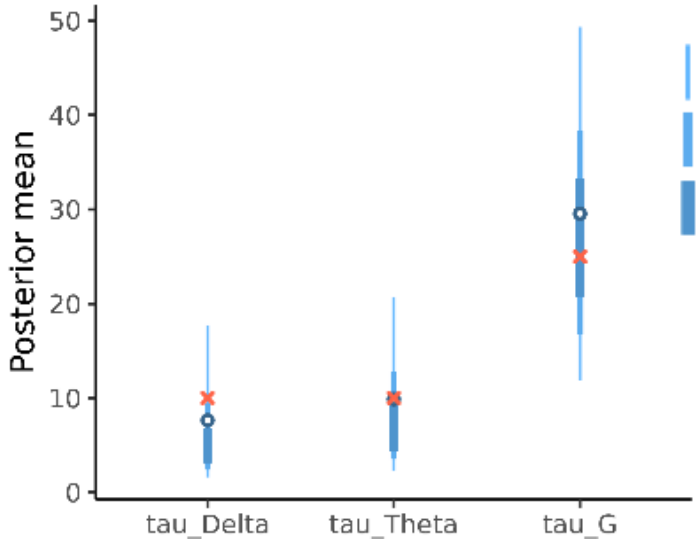


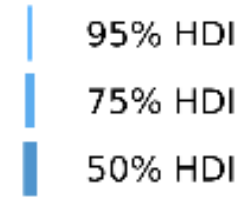




Guessing bias: G



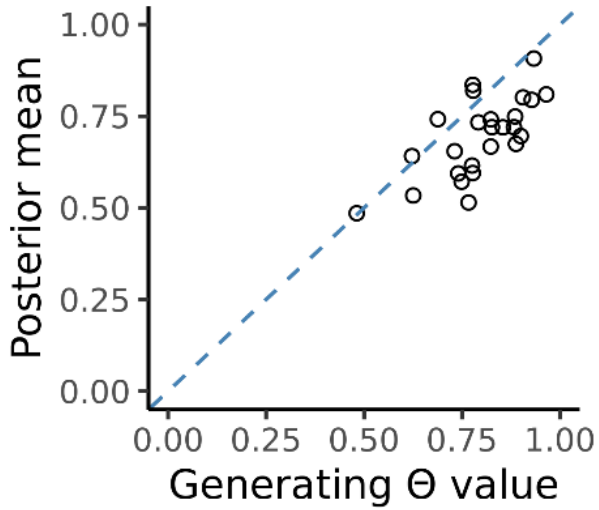

Hyperparameters

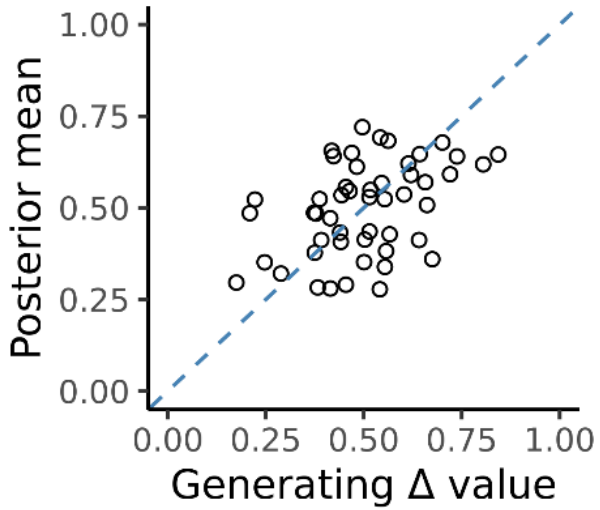


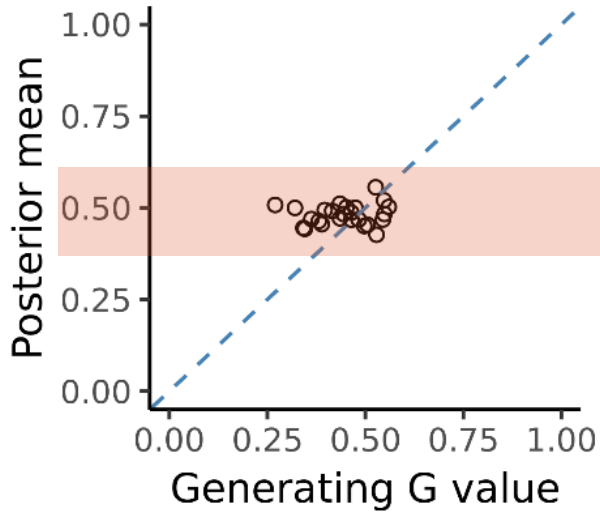



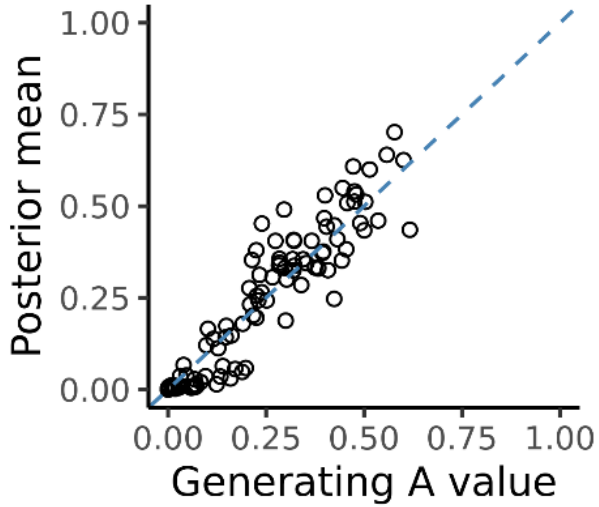

Hyperparameters

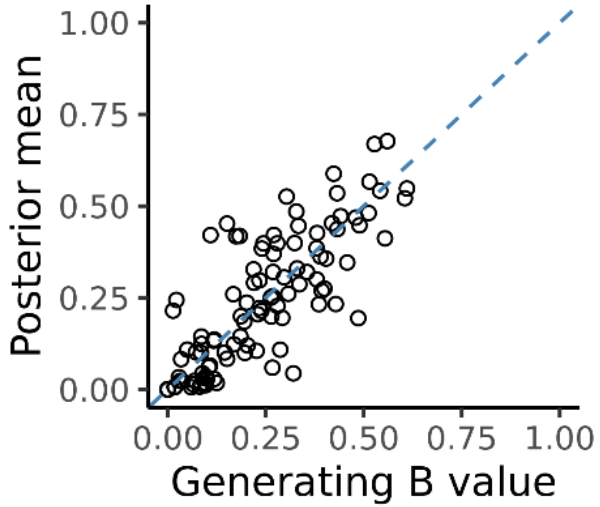
# Simulation results ($N = 25, M = 50, J = 5, V = 1$)




The guessing bias parameter ***G*** shows a strong shrinkage phenomenon ($cor = 0.119$).

# Parameters recovery

We simulated 96 conditions and ran each condition **50 times**

- # of informants: $N = 10, 25, 50, 100$
- # of items: $M = 10, 25, 50, 100$
- # of conf. ratings: $J = 2, 3, 4, 5, 6, 7$

Performance measurements:

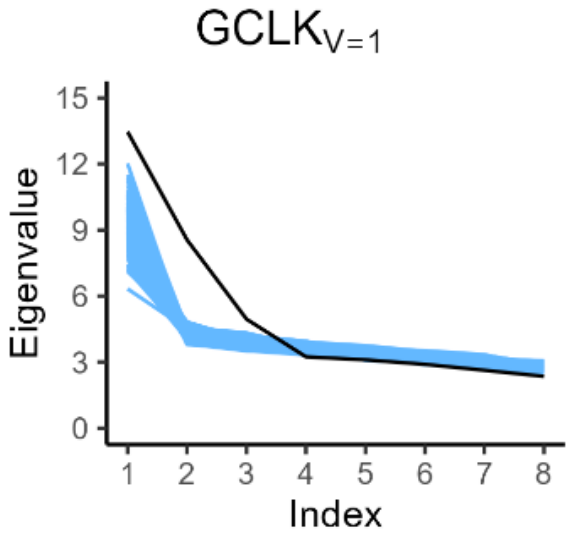
- Correlation
- Root mean square deviation
- 95% HDI recovery rate

The more response ($N \times M$) we have, the better the estimation performance is.

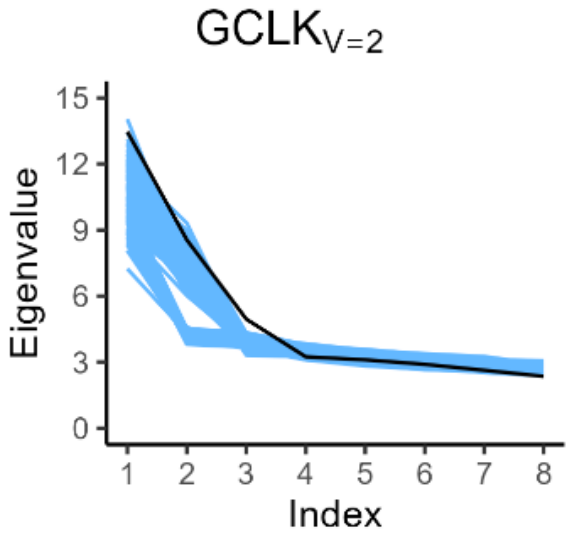
- Item-relative parameters, $Z_k$ and $\Delta_k$ were influenced more by $N$.
- Informant-relate parameters, $\theta_i$, $G_i$, $\boldsymbol{A_i}$, $\boldsymbol{B_i}$ were influenced more by $M$.
- However, $J$, does not help improve our estimation.

# Simulation results ($N = 75, M = 50, J = 5, V = 3$)

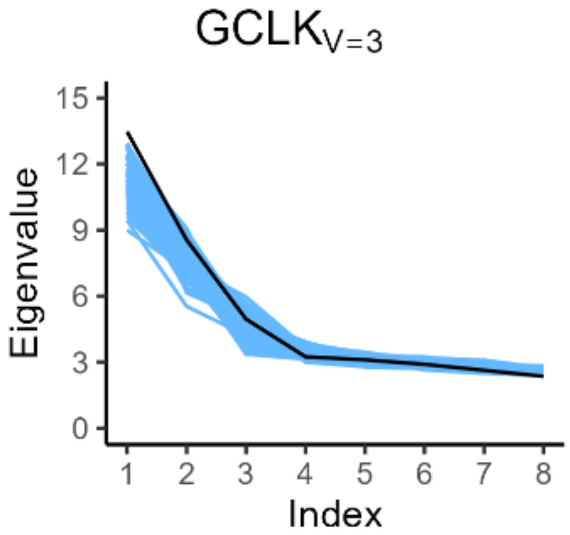
Eigenvalue posterior predictive check and the DIC



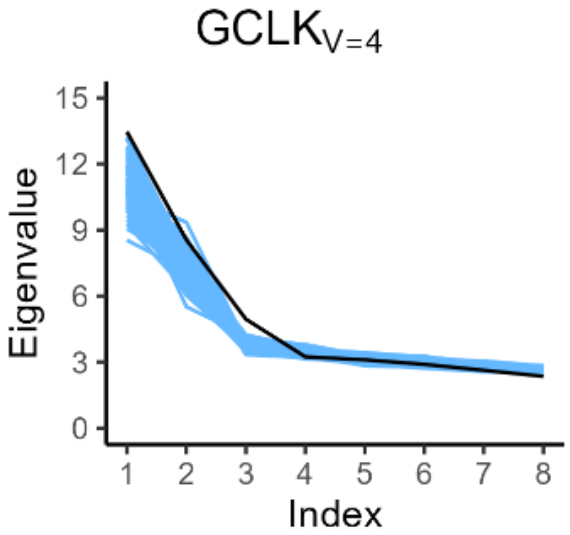

DIC = 11781.55




DIC = 11264.03



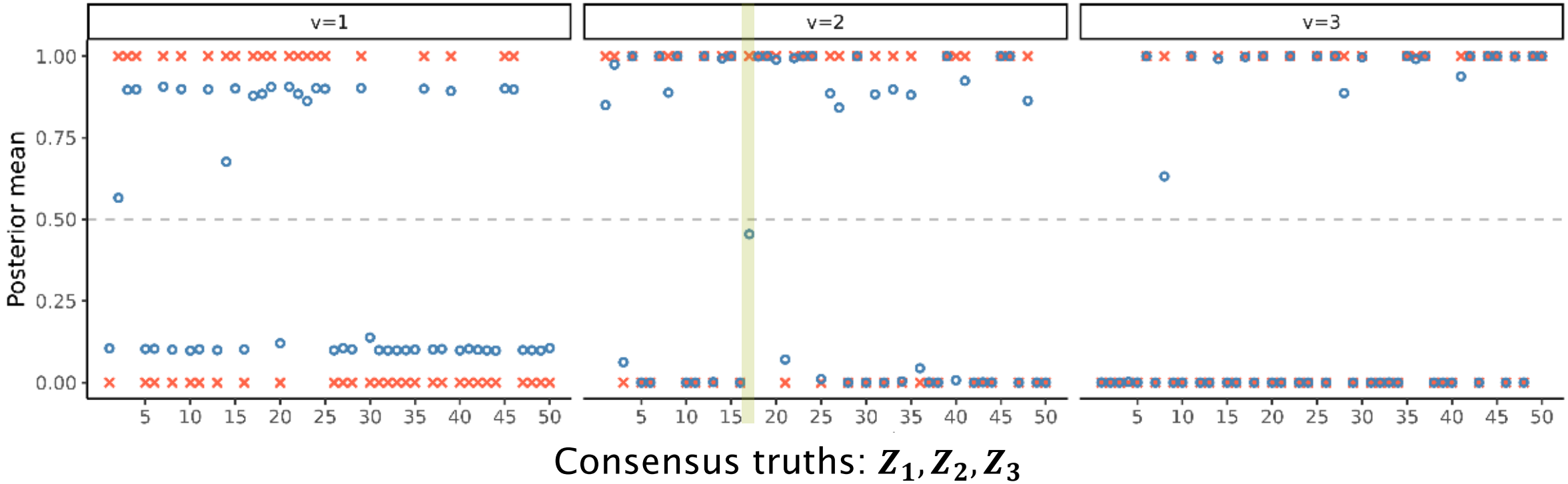

DIC = 10997.51




DIC = 11194.23

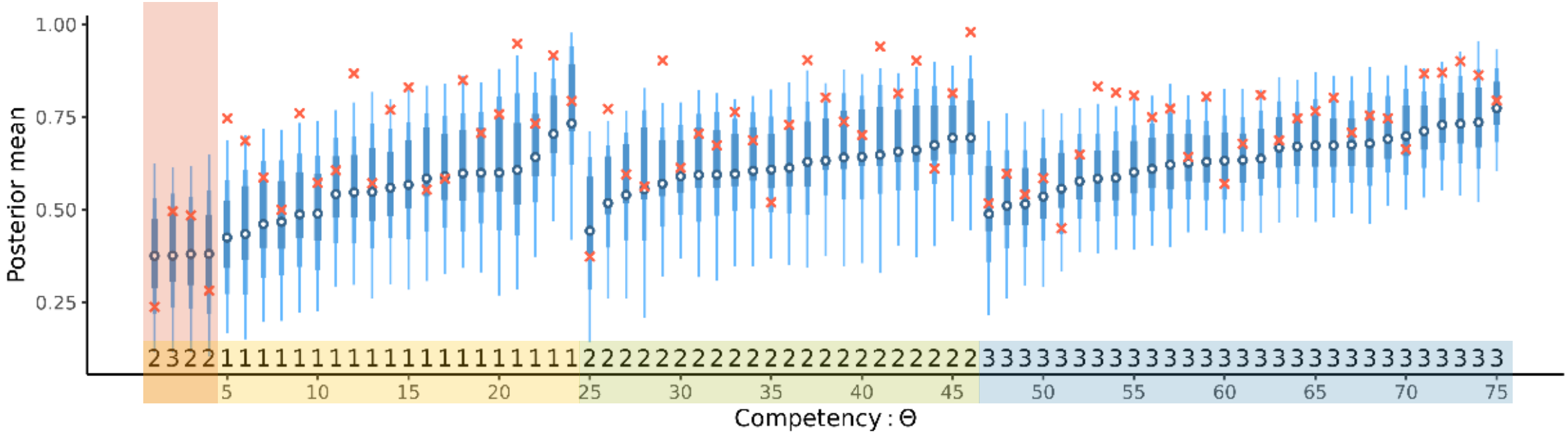
# Simulation results ($N = 75, M = 50, J = 5, V = 3$)

Consensus truths: $\mathcal{Z} = \{Z_1, Z_2, Z_3\}$



# Simulation results ($N = 75, M = 50, J = 5, V = 3$)

Informant's membership parameter $\Omega$ and competency $\Theta$




| | Predicted group | | |
|---|---|---|---|
| True group | 1 | 2 | 3 |
| 1 | **20** | 0 | 0 |
| 2 | 3 | **22** | 0 |
| 3 | 1 | 0 | **29** |

# Accuracy of Model Selection & Membership

- There was about 55% accuracy in selecting the correct model by the DIC.

| | $GCLK_{V=1}$ | $GCLK_{V=2}$ | $GCLK_{V=3}$ | $GCLK_{V=4}$ |
|---|---|---|---|---|
| $SIM_{V=1}$ | **12** | 1 | 5 | 2 |
| $SIM_{V=2}$ | 1 | **12** | 4 | 3 |
| $SIM_{V=3}$ | 0 | 3 | **11** | 6 |

- The average accuracy of consensus truth estimation was over 98%.
- The average accuracy of membership estimation was over 92%.

| | $GCLK_{V=1}$ for $SIM_{V=1}$ | $GCLK_{V=2}$ for $SIM_{V=2}$ | $GCLK_{=3}$ for $SIM_{V=3}$ |
|---|---|---|---|
| $\boldsymbol{Z} = \{\boldsymbol{Z}_1, \ldots, \boldsymbol{Z}_V\}$ | 0.994 (0.015) | 0.988 (0.016) | 0.988 (0.021) |
| $\boldsymbol{\Omega} = (\Omega_1, \ldots, \Omega_N)$ | - | 0.969 (0.056) | 0.928 (0.111) |

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Simulation Results

**General Discussion**

# Discussion

The **General Condorcet-Luce-Krantz (GCLK)** model extends

- the General Condorcet model + the two-state low-threshold model with confidence ratings
- the multicultural extension
- the hierarchical Bayesian model approach to model selection and parameter estimation.

In our simulation study, the GCLK expressed a strong recovery in

- model parameters and hyperparameters, number of the consensus cultures $V$ and informant's membership $\boldsymbol{\Omega}$.
- the more responses the data have, the well performances the estimates have.
- guessing bias $G$ showing a strong shrinkage phenomenon.
- the DIC seems to prefer the larger $v$-cultures model.

# Future Research

1. An empirical study of the GCLK is needed!

2. What kind of model selection criterion is suitable for the GCLK?
   - Alternative criteria: BPIC, WAIC, LOO-CV, or Bayes Factor

3. Explore the possibility of adding subject-level or culture-level covariates into the GCLK to explain the within- or between-groups variability.

4. Extend different kinds of CCT models, including the GCLK, into a generalized model.
   - A mixture of response formats: continuous + ordered categorical data + unordered categorical data (Aßfalg & Klauer, 2020)

# Q&A

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